PROJECT 2030 · MYAI FUTURE HACKATHON · TRACK 2 CITIZENS FIRST

Arus

arus (BM, n.) — the current of a river; the flow of people, information, and intent

AI that moves Malaysia's rescue faster than any human dispatcher can.

SOLO · LIU WEI · 2026-04-21

THE COST OF UNCOORDINATED RESCUE

148,024

Malaysians displaced

NADMA situation report · Dec 2024 · east-coast monsoon flood

100+

Orang Asli families cut off

Gua Musang interior, after rivers burst — manual relay only.

4

agencies, overlapping zones

BOMBA · NADMA · APM · MMEA — uncoordinated comms during the event.

"People die waiting for a human to add up the numbers."

THE PROBLEM IS NOT COURAGE. IT IS COORDINATION.

A dispatcher's job is not doable by humans.

What a human dispatcher tracks

- 5+ rescue assets across 20+ km
- Power / fuel budgets on each
- New triangulated calls every minute

What Arus does, in ~18 seconds / cycle

- 5-stage Gemini pipeline re-plans every cycle
- Power budgets enforced at the tool layer

THE TECHNICAL MOAT

MCP — the commander does not know the fleet.

Most agent demos hard-code `drone_ids`. Arus does not.

Gemini calls `tools/list_changed` every cycle and

```
┌─ Gemini 2.5 Flash ─┐
│                    │
│ tools/list_changed ──► MCP server (port 8001)
│                    │        │
│                    │        ├─ discover_fleet
│                    │        ├─ assign_search_mission
│                    │        ├─ list_detections  ⚡ Arus
│                    │        └─ ... 9 total
│                    │
│ Runtime tool bind  │
│ + JSON schemas     │
└────────────────────┘
```

GOOGLE ADK · SEQUENTIALAGENT

5 Gemini agents turn chaos into dispatch.

Each agent replaces a human role that saturates during surge.

STAGE 01 · REPLACES
Fleet Ops Officer

Assessor — reads fleet, coverage,

STAGE 02 · REPLACES
Tactical Planner

Strategist — scores targets, enforces power

STAGE 03 · REPLACES
Mission Controller

Dispatcher — commits missions,

FULL GOOGLE STACK — ZERO SHORTCUTS

Every mandated tool, used in its intended role.

Mandatory

Google Antigravity

Google AI Studio

Google Cloud Run (asia-southeast1)

```
<h3 class="mint" style="margin-top: 30px;">Google AI stack</h3>
<span class="pill">Gemini 2.5 Flash</span>
<span class="pill">Google ADK · SequentialAgent</span>
<span class="pill">Gemini Vision (JSON mode)</span>
```

Google Cloud

Firestore (asia-southeast1)

Artifact Registry

Secret Manager

Cloud Build

IAM + custom SA bindings

```
<h3 class="muted" style="margin-top: 30px;">Open protocols</h3>
<span class="pill">MCP (Model Context Protocol)</span>
<span class="pill">WebSocket / FastAPI</span>
```

A brief BOMBA can act on. In their language.

REAL OUTPUT · STAGE 5 · CYCLE 3

```
AGENSI: NADMA
KOORDINAT: (5, 9) — Kg. Aring, Gua Musang
KEUTAMAAN: TINGGI
RINGKASAN (BM): Mangsa dikesan di
  kawasan Kg. Aring, Gua Musang.
SUMMARY (EN): Victims detected in
  Kg. Aring area, Gua Musang.
CADANGAN TINDAKAN / RECOMMENDED ACTION:
  Penilaian lanjut dan tindakan
  menyelamat diperlukan.
  Further assessment and rescue
```

Why this wins

- BM for field teams whose first language is Malay.
- EN for NADMA coordinators writing federal reports.

FIELD FEATURE

Gemini Vision — photo in, agency out.

A BOMBA team phone uploads a drone frame to:

```
POST /api/vision/analyse
  multipart/form-data: file=roof.jpg
```

Gemini 2.5 Flash (JSON mode) returns:

- `victim_count` + `severity`
- `description_bm` + `description_en`

```
{
  "victim_count": 3,
  "severity": "CRITICAL",
  "description_bm":
    "Tiga individu terperangkap di atas bumbung.",
  "description_en":
    "Three individuals stranded on a roof.",
  "recommended_agency": "BOMBA",
  "hazards": ["fast current", "submerged debris"],
  "confidence": 0.92
}
```

MALAYSIA-INTEGRATED, NOT MALAYSIA-THEMED

Live MetMalaysia warnings feed reasoning.

Arus pulls real warnings from `api.data.gov.my/weather/warning` every cycle.

The Assessor sees them as part of its briefing.

The dashboard surfaces a MetMalaysia LIVE badge

TODAY · 2026-04-20 · 00:30 MYT

```
warning_issue:
  title_bm: Amaran Ribut Petir
  title_en: Thunderstorms Warning
  issued:   2026-04-20T00:30:00
valid_from:  2026-04-20T01:00:00
valid_to:    2026-04-20T05:00:00
text_en: Thunderstorms, heavy rain and

strong winds are expected over the

waters of Perlis & Kedah, Penang,
```

LIVE DEMO MOMENT

Watch the pipeline reason.

COMMANDCONSOLE · STREAMING · CYCLE 3

```
[assessor]   tool: get_situation_overview
             → fleet 5/5 · cov 48.8% · hotspot (5,9)
[assessor]   tool: list_detections
             → 4 victims · 2 Gua Musang · 2 Kota Tinggi
[strategist] tool: get_frontier_targets
             → SE quadrant 18.4% — focus here
[strategist] tool: plan_route Bravo → (5,9)
             → 7 cells · 14% · affordable ✓
[dispatcher] tool: assign_search_mission Bravo
             → accepted · ETA 7 ticks
[analyst]    SITREP built · Detections block OK
[agency]     emitted 4 BM/EN handoffs
```

Evidence layer

- Every tool call broadcast on WebSocket
- Every agent reasoning visible to operator
- No "trust me,

MALAYSIAN LOCALITY MODEL

A map commanders already know.

Arus does not report "Sector NE-3".

Arus reports Kg. Aring, Gua Musang.

That matters because:

- Dispatchers already have kampung phone trees
- BOMBA Balai names map

Current coverage

Gua Musang, Kelantan

Kuala Krai, Kelantan

Kota Tinggi, Johor

Segamat, Johor

```
<p class="muted" style="margin-top: 24px; font-size: 16px;">
  Demo Day is at UTM Skudai (Johor) — Dec 2024 flood localities are the freshest reference for the judges in the room.
</p>
<p class="muted" style="margin-top: 12px; font-size: 16px;">
  Kampung naming deterministic — same grid cell always maps to the same kampung across runs.
</p>
```

IMPACT MODEL

Minutes become lives.

~45 min

current dispatch lag

Manual phone-tree, multi-agency, per remote kampung.

~18 sec

Arus cycle latency

Full pipeline: MetMalaysia fetch + 5-stage Gemini + MCP dispatch.

150× faster

dispatch decision loop

Each saved minute in flood water is ~2-4 % survival-probability

SCALE PATH

Same codebase, new geographies.

Now

Peninsula east-coast monsoon belt · Kelantan + Johor.

- 4 districts
- 16 kampungs

+1 week

Sabah (Beaufort, Papar) + Sarawak (Kapit, Mukah) river floods.

- Swap `locality.py`

+1 month

ASEAN monsoon belt · Philippines typhoons, Vietnam Mekong floods, Thailand Chao Phraya

NATIONAL ALIGNMENT

Arus speaks Kerajaan's language.

Malaysia Madani

Digital-transformation of public safety is a cited priority. Arus is agency-collaborative by design.

```
<h3 class="cyan" style="margin-top: 30px;">MyDIGITAL</h3>
<p style="font-size: 20px;" class="muted">"AI for the Rakyat" — Arus applies frontier AI to a mass-market public-safety problem, not enterprise productivity.</p>

<h3 class="cyan" style="margin-top: 30px;">AI Nation 2030</h3>
<p style="font-size: 20px;" class="muted">Demonstrates Malaysian builders shipping production agentic systems on the Google stack, in Malaysia's language.</p>
```

GovTech Malaysia Unit

- Multi-agency workflow
- Bilingual by default
- Firestore-backed audit log (§ records retention)
- Cloud Run sovereign region (`asia-southeast1`)
- Zero vendor lock beyond Google AI

Arus

AI that moves Malaysia's rescue — faster than any human dispatcher can.

github.com/SunflowersLwtech/arus

arus-1030181742799.asia-southeast1.run.app

Built solo by Liu Wei · Project 2030 · MyAI Future Hackathon · Track 2 · 2026-04-21

Terima kasih. Thank you. .